

General Relativity as Transport Data in Resolution Geometry

Resolving the Lorentz-Signature Obstruction without Identifying the GR Metric with the Observation Metric

Matěj Rada

July 23, 2026

Abstract

Resolution Geometry (RG) requires a positive-definite observation metric in order to define a clean resolved/null decomposition. General relativity (GR), however, is formulated on a Lorentzian spacetime metric. This note states the resulting signature obstruction precisely and gives the surviving placement of GR inside RG. The GR metric is *not* inserted into the RG metric slot, and the stress-energy tensor is *not* identified with a response Jacobian. Instead, GR supplies Levi-Civita transport. Given an observation model, this transport acts on the resolved projector and produces an infinitesimal resolved–null mixing block

$$A_{RN} = P_N(\nabla^{LC} P_R)P_R.$$

This block is not itself a groupoid arrow. It is the off-diagonal component of a connection. Its path-ordered exponential gives adapted transport between local resolution geometries; closed loops yield automorphisms only when the transported resolved/null/covariance structure closes. Schwarzschild equatorial-loop holonomy supplies a concrete GR input: the holonomy fraction is

$$\beta(r) = 1 - \sqrt{1 - 2M/r},$$

which induces nonzero resolved–null mixing while the RG observation metric remains positive-definite. We report a battery that confirms the standard geometric hierarchy

$$A_{RN} \longrightarrow \text{connection} \longrightarrow \text{transport groupoid} \longrightarrow \text{closed-loop holonomy} / \text{obstruction}.$$

The result is a placement theorem, not a derivation of GR, not a derivation of topology, and not an identification of the reactive signature with spacetime.

All GR structures remain in their usual spacetime role; RG only uses the Levi-Civita connection as transport data acting on an externally chosen observation model F .

Contents

1	The problem	2
2	GR input data	2
3	The infinitesimal coupling	3
4	Transport and groupoid level	3
5	Schwarzschild equatorial-loop holonomy as GR input	4

6	Reactive layer and signature	4
7	Final bridge battery	5
8	Open issues	5
9	Non-claims	6
10	Conclusion	6
A	Numerical verification of the Schwarzschild equatorial-loop holonomy	6
A.1	Parallel-transport ODE	6
A.2	Reproducibility script	7
A.3	Numerical results	8
A.4	Interpretation	8

1 The problem

RG starts from local observation geometry

$$\mathcal{G}_x = (O_x, g_{\text{obs}x}; R_x, \Sigma_x), \quad R_x = \text{Im}(J_x), \quad N_x = R_x^{\perp g_{\text{obs}}}.$$

The metric g_{obs} must be positive-definite if the resolved/null split is to behave as a direct orthogonal decomposition. GR instead supplies a Lorentzian metric g_{GR} . On a Lorentzian vector space, null vectors may satisfy $v \in v^\perp$, so replacing g_{obs} by g_{GR} destroys the clean RG split. This is the Lorentz-signature obstruction.

The surviving placement is therefore not

$$g_{\text{GR}} \mapsto g_{\text{obs}},$$

and not

$$T_{\mu\nu} \mapsto J.$$

Rather, GR enters through transport:

$$(M, g_{\text{GR}}, \nabla^{\text{LC}}) \text{ acts on } P_R.$$

The observation metric remains positive-definite; Lorentzian GR data are represented by the Levi-Civita connection and its holonomy.

2 GR input data

Definition 1 (GR-to-RG input package). A GR observation package consists of

$$(M, g_{\text{GR}}, \nabla^{\text{LC}}; F, \Sigma),$$

where $(M, g_{\text{GR}}, \nabla^{\text{LC}})$ is a Lorentzian spacetime with Levi-Civita connection, F is an observation map, and Σ is the positive-definite observation covariance defining $g_{\text{obs}} = \Sigma^{-1}$ or another declared positive-definite observation metric.

The map F is not a defect: GR alone does not define what is observed. Different instruments define different response maps. Once F is fixed, $J = dF$ and hence P_R are fixed in the local observation geometry.

3 The infinitesimal coupling

Definition 2 (Resolved-null coupling block). Let P_R and $P_N = I - P_R$ denote the positive-definite RG projectors. The GR-induced off-diagonal coupling is

$$A_{RN} = P_N(\nabla^{\text{LC}} P_R)P_R.$$

It is an infinitesimal connection component, not an automorphism.

Gauge changes of resolved and null frames transform

$$A_{RN} \mapsto V_N^{-1} A_{RN} U_R.$$

Thus raw matrix entries are frame-dependent, while quantities such as the singular values of A_{RN} or the spectrum of $A_{RN}^* A_{RN}$ are gauge-invariant resolved-null mixing data.

4 Transport and groupoid level

Definition 3 (Adapted RG transport). Let $\gamma : [0, 1] \rightarrow M$ be a path. A transport $\mathcal{T}_\gamma : \mathcal{O}_{\gamma(0)} \rightarrow \mathcal{O}_{\gamma(1)}$ is adapted if it maps the full RG object at the initial point to the transported RG object at the endpoint:

$$\mathcal{T}_\gamma R_{\gamma(0)} = R_{\gamma(1)}^\gamma, \quad \mathcal{T}_\gamma N_{\gamma(0)} = N_{\gamma(1)}^\gamma, \quad \mathcal{T}_\gamma^* \Sigma_{\gamma(1)}^\gamma = \Sigma_{\gamma(0)}.$$

Theorem 1 (Transport-to-groupoid principle). *If A is an RG-compatible connection preserving the positive-definite observation metric and transporting the pair (R, N) as a split object, then*

$$\mathcal{T}_\gamma = \mathcal{P} \exp \int_\gamma A$$

defines a morphism in the transport groupoid of local resolution geometries. Composition of paths corresponds to composition of transports.

Proof. This is the standard proof for parallel transport of a compatible connection. Metric compatibility gives

$$\frac{d}{dt} (\mathcal{T}_t^* \Sigma_t \mathcal{T}_t) = 0,$$

so the observation covariance is preserved. Compatibility with the moving split gives $\mathcal{T}_t R_0 = R_t$ and $\mathcal{T}_t N_0 = N_t$. Uniqueness of ODE solutions for parallel transport yields $\mathcal{T}_{\gamma_2 \circ \gamma_1} = \mathcal{T}_{\gamma_2} \mathcal{T}_{\gamma_1}$ whenever paths compose. Hence the transports form groupoid morphisms between RG objects. $\square$

Proposition 1 (Closed-loop automorphism criterion). *For a closed base loop γ , the transport $\mathcal{T}_\gamma$ belongs to $\text{Aut}(\mathcal{G}_x)$ if and only if the transported resolution structure closes:*

$$\mathcal{T}_\gamma R_x = R_x, \quad \mathcal{T}_\gamma N_x = N_x, \quad \mathcal{T}_\gamma^* \Sigma_x = \Sigma_x.$$

If the base loop closes but the resolved projector returns rotated, then $\mathcal{T}_\gamma$ is not an automorphism of the original object; it is a monodromy defect, namely an arrow from $\mathcal{G}_x$ to the transported object $\mathcal{G}_x^\gamma$.

5 Schwarzschild equatorial-loop holonomy as GR input

We use the Schwarzschild metric in units $G = c = 1$,

$$ds^2 = -f(r) dt^2 + f(r)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad f(r) = 1 - \frac{2M}{r}.$$

Consider the closed loop

$$\gamma_r(\phi) = (t_0, r, \pi/2, \phi), \quad \phi \in [0, 2\pi],$$

at fixed $t = t_0$, fixed radius $r > 2M$, and in the equatorial plane. For a vector transported in the (r, ϕ) -sector, the Levi-Civita parallel-transport equations along this loop reduce to

$$\frac{dV^r}{d\phi} = f(r) r V^\phi, \quad \frac{dV^\phi}{d\phi} = -\frac{1}{r} V^r.$$

Consequently,

$$\frac{d^2 V^r}{d\phi^2} = -f(r) V^r,$$

so the transported vector rotates in the local orthonormal $(\hat{r}, \hat{\phi})$ -frame with angular frequency $\sqrt{f(r)}$. After one coordinate loop, the within-frame rotation is

$$2\pi \sqrt{f(r)} = 2\pi \sqrt{1 - \frac{2M}{r}}.$$

The corresponding deficit angle, or loop holonomy angle, is therefore

$$\Delta\alpha(r) = 2\pi - 2\pi \sqrt{1 - \frac{2M}{r}} = 2\pi \left(1 - \sqrt{1 - \frac{2M}{r}} \right).$$

We define the dimensionless holonomy fraction

$$\boxed{\beta(r) = \frac{\Delta\alpha(r)}{2\pi} = 1 - \sqrt{1 - \frac{2M}{r}}}.$$

This $\beta(r)$ is a genuine Levi-Civita parallel-transport holonomy for the selected equatorial circular loop, not a hand-chosen Schwarzschild-scaled rate. It is path-dependent, as holonomy must be: other loops in Schwarzschild spacetime generally produce different transport maps. In RG it is used only as a GR transport datum feeding the projector coupling

$$A_{RN} = P_N(\nabla^{LC} P_R) P_R.$$

It does not replace the positive-definite RG observation metric by the Lorentzian spacetime metric.

6 Reactive layer and signature

The RG observation metric remains positive-definite. The derived reactive form is a different object. With a passive null-sector model,

$$K(\tau) = A_{RN}^* e^{A_N \tau} A_{RN},$$

and in frequency domain

$$Q(\omega) = K_R - \omega^2 M - \text{Herm} \Sigma_R(\omega).$$

Since the self-energy is quadratic in A_{RN} , small Schwarzschild holonomy gives wall shifts of order

$$\Delta\omega_i = O(-\beta(r)^2)$$

under stable passive null dynamics. This is a reactive signature effect, not an identification with the spacetime metric.

7 Final bridge battery

We tested the final hierarchy on $O = \mathbb{R}^4$ with positive-definite metric, $\dim R = \dim N = 2$, and Schwarzschild-loop input $r = 8M$:

$$\beta = 0.1339745962, \quad A_{RN} = \text{diag}(0.1339745962, 0.0602885683).$$

Test	Outcome
Adapted open arrow	Orthogonality error 1.47×10^{-16} ; maps initial R, N to endpoint transported R, N with errors $< 10^{-16}$.
Closed base loop, generic split	Orthogonal, but does not fix the initial split: R -fixed error 1.055, N -fixed error 1.055. Thus it is not generally in $\text{Aut}(\mathcal{G}_x)$.
Groupoid composition	Segmented transports compose with error 1.39×10^{-16} .
Closed automorphism case	When the transported split closes, fixed-split errors drop to machine precision.
Naive A_{RN} only	Exponentiating an off-diagonal block without compatible completion fails metric preservation; metric error 1.78.
Skew completion	Metric preservation succeeds, but nonzero off-diagonal mixing does not preserve a fixed split; it is transport between split objects, not an automorphism of one object.

Result 1 (Surviving bridge statement). *GR data stand correctly in RG when placed as Levi-Civita transport of the observation projector:*

$$(M, g, \nabla^{\text{LC}}; F, \Sigma) \implies (O, g_{\text{obs}}; R, N, \Sigma) \implies A_{RN} = P_N(\nabla^{\text{LC}} P_R) P_R.$$

The block A_{RN} is the infinitesimal resolved-null mixing component. Its compatible path-ordered exponential gives arrows of the RG transport groupoid. Closed loops give automorphisms precisely when the transported RG object closes; otherwise they give monodromy defects.

8 Open issues

1. **Null-sector dynamics.** The size and sign of the reactive wall shift require admissibility assumptions on A_N . Stable passive dynamics give a robust downward shift; active/unstable null dynamics can reverse it.
2. **Observation model dependence.** GR does not define F . The observation model is necessary input. The theorem says how GR transport acts once the observation geometry has been chosen.
3. **Global closure.** Local transport does not determine global topology. Closed-loop automorphism requires closure of (R, N, Σ) ; failure to close is monodromy, not contradiction.
4. **Reactive signature ceiling.** The derived Lorentzian band of $Q(\omega)$ is a response signature, not spacetime, not curvature, and not Einstein dynamics.

9 Non-claims

Non-claim 1. No derivation of GR from RG or QM is claimed.

Non-claim 2. The Lorentzian spacetime metric is not placed in the positive-definite RG metric slot.

Non-claim 3. The stress-energy tensor is not identified with a Jacobian or Fisher matrix.

Non-claim 4. A_{RN} alone is not a groupoid arrow and does not determine full holonomy. It is an infinitesimal off-diagonal connection component.

Non-claim 5. Schwarzschild holonomy is path-dependent. The formula $\beta(r) = 1 - \sqrt{1 - 2M/r}$ refers to the selected equatorial circular loop.

10 Conclusion

The Lorentz-signature obstruction is resolved by separating layers. The RG observation metric remains positive-definite. GR enters through Levi-Civita transport, not by replacing the RG metric or by identifying stress-energy with response. The off-diagonal term A_{RN} is the infinitesimal generator of resolved-null leakage. Integrated compatible transports give groupoid arrows between resolution geometries; closed-loop automorphisms occur exactly when the transported split/covariance structure closes. This is the admissible geometric placement of GR data inside Resolution Geometry.

A Numerical verification of the Schwarzschild equatorial-loop holonomy

This appendix verifies the formula

$$\beta(r) = 1 - \sqrt{1 - \frac{2M}{r}}$$

by direct numerical integration of the Levi-Civita parallel-transport equations along the fixed- t , fixed- r , equatorial circular loop.

A.1 Parallel-transport ODE

For the loop

$$\gamma_r(\phi) = (t_0, r, \pi/2, \phi), \quad \phi \in [0, 2\pi],$$

the relevant transport equations in the (r, ϕ) -sector are

$$\frac{dV^r}{d\phi} = frV^\phi, \quad \frac{dV^\phi}{d\phi} = -\frac{1}{r}V^r, \quad f = 1 - \frac{2M}{r}.$$

Starting with an initially radial vector,

$$V^r(0) = 1, \quad V^\phi(0) = 0,$$

we integrate the ODE numerically. The final coordinate components are converted to orthonormal components by

$$V^{\hat{r}} = \frac{V^r}{\sqrt{f}}, \quad V^{\hat{\phi}} = rV^\phi.$$

The accumulated angle in the orthonormal frame is computed as

$$\alpha_{\text{num}} = \text{atan2}(-V^{\hat{\phi}}, V^{\hat{r}}).$$

Modulo the orientation convention, this agrees with

$$\alpha_{\text{an}} = 2\pi\sqrt{f} \pmod{2\pi}.$$

The holonomy angle is then

$$\Delta\alpha_{\text{num}} = 2\pi - \alpha_{\text{num}} \pmod{2\pi},$$

and should agree with

$$\Delta\alpha_{\text{an}} = 2\pi(1 - \sqrt{f}).$$

A.2 Reproducibility script

```
import numpy as np
from scipy.integrate import solve_ivp

def wrap(a):
    # wrap angle to (-pi, pi]
    return (a + np.pi) % (2*np.pi) - np.pi

def holonomy_check(r, M=1.0):
    f = 1 - 2*M/r
    sf = np.sqrt(f)

    def rhs(phi, V):
        Vr, Vphi = V
        return [f*r*Vphi, -(1/r)*Vr]

    sol = solve_ivp(
        rhs,
        [0, 2*np.pi],
        [1.0, 0.0],
        rtol=1e-12,
        atol=1e-14
    )

    Vr, Vphi = sol.y[:, -1]

    # Orthonormal components
    Vrh = Vr / np.sqrt(f)
    Vphih = r * Vphi

    # Numerical within-frame angle
    angle_numeric = np.arctan2(-Vphih, Vrh)

    # Analytic within-frame angle, modulo 2pi
    angle_analytic = wrap(2*np.pi*sf)

    # Holonomy angle
```

```

holo_numeric = wrap(2*np.pi - angle_numeric)
holo_analytic = wrap(2*np.pi*(1 - sf))

return {
    "r_over_M": r,
    "sqrt_f": sf,
    "angle_error": abs(wrap(angle_numeric - angle_analytic)),
    "holonomy_error": abs(wrap(holo_numeric - holo_analytic)),
    "beta": 1 - sf,
    "holonomy_over_2pi": (2*np.pi*(1 - sf))/(2*np.pi)
}

for r in [6, 8, 12, 20, 50]:
    print(holonomy_check(r))

```

A.3 Numerical results

The following table reports the numerical agreement between the direct ODE integration and the analytic holonomy formula.

r/M	$\sqrt{f}$	$\beta(r) = 1 - \sqrt{f}$	$\Delta\alpha/(2\pi)$	holonomy error
6	0.8164965809	0.1835034191	0.1835034191	1.4×10^{-14}
8	0.8660254038	0.1339745962	0.1339745962	1.7×10^{-14}
12	0.9128709292	0.0871290708	0.0871290708	1.8×10^{-14}
20	0.9486832981	0.0513167019	0.0513167019	2.0×10^{-14}
50	0.9797958971	0.0202041029	0.0202041029	2.8×10^{-14}

Thus direct numerical Levi-Civita parallel transport confirms

$$\frac{\Delta\alpha}{2\pi} = 1 - \sqrt{1 - \frac{2M}{r}}$$

to numerical precision.

A.4 Interpretation

The computation verifies only the selected loop:

$$\gamma_r(\phi) = (t_0, r, \pi/2, \phi).$$

It is not a claim about all Schwarzschild holonomies and not a claim that $\beta(r)$ is a scalar invariant of the spacetime independent of path. It is a path-dependent Levi-Civita transport datum. Its role in RG is to provide a concrete GR-induced transport input for

$$A_{RN} = P_N(\nabla^{LC} P_R)P_R,$$

while the RG observation metric remains positive-definite.

References

- [1] T. Rothman, G. F. R. Ellis, and J. Murugan, “Holonomy in the Schwarzschild-Droste Geometry,” *Classical and Quantum Gravity* 18 (2001), 1217–1234; arXiv:gr-qc/0008070.
- [2] K. Sakina and J. Chiba, “Parallel transport of a vector along a circular orbit in Schwarzschild space-time,” *Physical Review D* 19 (1979), 2280.

- [3] R. Zwanzig, “Ensemble method in the theory of irreversibility,” *Journal of Chemical Physics* 33 (1960), 1338–1341.
- [4] H. Mori, “Transport, collective motion, and Brownian motion,” *Progress of Theoretical Physics* 33 (1965), 423–455.
- [5] J. J. Sylvester, “A demonstration of the theorem that every homogeneous quadratic polynomial is reducible by real orthogonal substitutions to the form of a sum of positive and negative squares,” *Philosophical Magazine* 4 (1852), 138–142.
- [6] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vols. I–II, Wiley, 1963–1969.